Qualification Pack

Artificial Intelligence Assistant

QP Code: NIE/SSC/Q1003

Version: 1.0

NSQF Level: 3

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Contents

NIE/SSC/Q1003: Artificial Intelligence Assistant

Brief Job Description

An Artificial Intelligence Assistant facilitates user interactions with AI systems, providing support, guidance, and executing tasks based on user requests, enhancing productivity and efficiency through intelligent automation and natural language processing.

Personal Attributes

The job of an Artificial Intelligence Assistant requires strong communication skills, adaptability, and a keen understanding of both user needs and emerging AI technologies to effectively facilitate human-AI interactions and deliver personalized assistance.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. NIE/ITS/N1001: PROGRAMMING WITH PYTHON
2. NIE/ITS/N1002: CONCEPTUALISING DATA SCIENCE WITH PYTHON
3. NIE/ITS/N1003: DATA ANALYSIS AND VISUALIZATION
4. NIE/ITS/N1004: FUNDAMENTALS OF MACHINE LEARNING
5. DGT/VSQ/N0101: Employability Skills (30 Hours)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Artificial Intelligence
Occupation	AI Development
Country	India
NSQF Level	3
Credits	10
Aligned to NCO/ISCO/ISIC Code	2511.0106: Support Engineer

Minimum Educational Qualification & Experience	10th grade pass OR 8th grade pass plus 2-year NTC plus 1 Year NAC OR 8th grade pass and pursuing continuous schooling in regular school with vocational subject OR 8th grade pass with 2 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	30/06/2026
NSQC Approval Date	29/03/2023
Version	1.0
Reference code on NQR	QG-03-IT-00356-2023-V1-NIELIT
NQR Version	1

NIE/ITS/N1001: Programming with Python

Description

Python Programming cover the installation and configuration of a Python environment, basic programming concepts, data structures, and the use of modules and packages to provide a foundational understanding of Python development.

Scope

The scope covers the following :

- IDE Installation, Programming, Exploring Data Structures

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Learn to install and configure the python IDE and learn about working on the collaborative cloud interface required for programming
PC2. Understand the basics of Python Language and will be able to recognize Python syntax
PC3. Understand and write programs in python language, compile, debug and handle exceptions
PC4. Working with top down or bottom up approach by making functions.
PC5. Do modular programming by defining and calling various functions
PC6. Working with different data structures which allow to organize and store data
PC7. Apply algorithms to process data in various data structures in a meaningful way

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Python Environment Setup: The student will be able to set up a Python programming environment, including installing Python and configuring essential tools like Integrated Development Environments (IDEs) or code editors.
KU2. Basic Programming Skills: The student will have a solid grasp of fundamental programming concepts, such as variables, data types, operators, conditional statements, loops, and functions.
KU3. Datatypes and Operators: They will understand how to work with various data types (e.g., integers, strings, lists) and apply operators (e.g., arithmetic, comparison, logical) to manipulate data.
KU4. Control Flow: The student will be able to create programs with control flow structures, including conditional statements (if, else, and elif) and loops (for and while) to control program execution.
KU5. Functions: They will know how to define and use functions, including passing arguments, returning values, and organizing code for reusability.

KU6. Data Structures: The student will gain knowledge about different data structures like lists, dictionaries, sets, and tuples, and understand how to select the appropriate data structure for specific tasks.

KU7. Modules and Packages: They will learn how to create, import, and use Python modules and packages to organize and manage code efficiently, promoting code modularity and reusability.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Students will develop general skills like problem-solving, logical thinking, attention to detail, and communication.

GS2. Students will also acquire professional skills, including programming proficiency, code efficiency, modular programming, adaptability, and knowledge of software development best practices.

GS3. This comprehensive skill set equips students for diverse career opportunities in technology and programming, providing a strong foundation for their professional development

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	**40**	**-**	**-**
PC1. Learn to install and configure the python IDE and learn about working on the collaborative cloud interface required for programming	2	4	-	-
PC2. Understand the basics of Python Language and will be able to recognize Python syntax	3	6	-	-
PC3. Understand and write programs in python language, compile, debug and handle exceptions	3	6	-	-
PC4. Working with top down or bottom up approach by making functions.	3	6	-	-
PC5. Do modular programming by defining and calling various functions	3	6	-	-
PC6. Working with different data structures which allow to organize and store data	3	6	-	-
PC7. Apply algorithms to process data in various data structures in a meaningful way	3	6	-	-
NOS Total	**20**	**40**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1001
NOS Name	Programming with Python
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

NIE/ITS/N1001: PROGRAMMING WITH PYTHON

Description

Python Programming cover the installation and configuration of a Python environment, basic programming concepts, data structures, and the use of modules and packages to provide a foundational understanding of Python development.

Scope

The scope covers the following :

- IDE Installation, Programming, Exploring Data Structures

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Learn to install and configure the python IDE and learn about working on the collaborative cloud interface required for programming

PC2. Understand the basics of Python Language and will be able to recognize Python syntax

PC3. Understand and write programs in python language, compile, debug and handle exceptions

PC4. Working with top down or bottom up approach by making functions.

PC5. Do modular programming by defining and calling various functions

PC6. Working with different data structures which allow to organize and store data

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Python Environment Setup: The student will be able to set up a Python programming environment, including installing Python and configuring essential tools like Integrated Development Environments (IDEs) or code editors.

KU2. Basic Programming Skills: The student will have a solid grasp of fundamental programming concepts, such as variables, data types, operators, conditional statements, loops, and functions.

KU3. Datatypes and Operators: They will understand how to work with various data types (e.g., integers, strings, lists) and apply operators (e.g., arithmetic, comparison, logical) to manipulate data.

KU4. Control Flow: The student will be able to create programs with control flow structures, including conditional statements (if, else, and elif) and loops (for and while) to control program execution.

KU5. Functions: They will know how to define and use functions, including passing arguments, returning values, and organizing code for reusability.

KU6. Data Structures: The student will gain knowledge about different data structures like lists, dictionaries, sets, and tuples, and understand how to select the appropriate data structure for specific tasks.

KU7. Modules and Packages: They will learn how to create, import, and use Python modules and packages to organize and manage code efficiently, promoting code modularity and reusability.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Students will develop general skills like problem-solving, logical thinking, attention to detail, and communication.

GS2. Students will also acquire professional skills, including programming proficiency, code efficiency, modular programming, adaptability, and knowledge of software development best practices.

GS3. This comprehensive skill set equips students for diverse career opportunities in technology and programming, providing a strong foundation for their professional development

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	**22**	**-**	**-**
PC1. Learn to install and configure the python IDE and learn about working on the collaborative cloud interface required for programming	5	2	-	-
PC2. Understand the basics of Python Language and will be able to recognize Python syntax	10	4	-	-
PC3. Understand and write programs in python language, compile, debug and handle exceptions	10	4	-	-
PC4. Working with top down or bottom up approach by making functions.	10	4	-	-
PC5. Do modular programming by defining and calling various functions	10	4	-	-
PC6. Working with different data structures which allow to organize and store data	5	4	-	-
NOS Total	**50**	**22**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1001
NOS Name	PROGRAMMING WITH PYTHON
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	3
Credits	2
Version	1.0
Next Review Date	NA

NIE/ITS/N1002: Problem-solving and algorithmic thinking using the knowledge of Python libraries and frameworks

Description

This NOS focuses on developing problem-solving skills and algorithmic thinking while leveraging Python libraries and frameworks to tackle real-world challenges, equipping students with practical programming expertise.

Scope

The scope covers the following :

- Optimizing Data Processing, Data Science and Analysis

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Ability to employ problem-solving and algorithmic thinking to efficiently handle and manipulate data using Python libraries such as NumPy and Pandas. Utilize these tools to clean, transform, and analyze data, enabling better decision-making and insights.

PC2. Ability to utilize Python libraries such as Pandas, NumPy, and Matplotlib to solve intricate data analysis problems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. • A.Organizational
• Context
• KA1. Efficiency Enhancement
• KA2.Innovation and Solution Development
• KA3. Cross-functional Collaboration

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. • SA1.Fill up documentation applicable to ones role.
• SA2. Read English and/or vernacular language.
• SA3. Read and understand manuals, health and safety instructions, memos, other company documents.
• SA4. Ability to read from different sources- books screens in machines and signage.
• SA5. Understand the various color codes, as per standard electrical,mechanical
• SA6. Express statements or information clearly so that others can hear and understand.

GS2. • SB1. Follow organization rule-based decision making process.
• SB2. Take decision with systematic course of actions and/or response.
• The user/individual on the job needs to know and understand how to :
• SB3. Planning and organization of work to meet deadlines.
• SB4. Work constructively and collaboratively with others.
• SB5. Follow code of conduct.
• SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
• SB7. Recognize problems and search for solutions.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	**12**	**-**	**6**
PC1. Ability to employ problem-solving and algorithmic thinking to efficiently handle and manipulate data using Python libraries such as NumPy and Pandas. Utilize these tools to clean, transform, and analyze data, enabling better decision-making and insights.	10	6	-	3
PC2. Ability to utilize Python libraries such as Pandas, NumPy, and Matplotlib to solve intricate data analysis problems.	10	6	-	3
NOS Total	**20**	**12**	**-**	**6**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1002
NOS Name	Problem-solving and algorithmic thinking using the knowledge of Python libraries and frameworks
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence , Data Science & Engineering
Occupation	AI Development
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

NIE/ITS/N1002: CONCEPTUALISING DATA SCIENCE WITH PYTHON

Description

This NOS provides a comprehensive introduction to data science concepts and techniques using the Python programming language. Students learn how to analyze data, build predictive models, and gain a foundational understanding of data science principles.

Scope

The scope covers the following :

- Data Analysis and Visualization, Machine Learning Fundamentals, Conceptual Understanding of Data Science

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Students should be able to effectively explore and analyze datasets using Python, including data cleaning, feature engineering, and visualization.

PC2. Students should exhibit a solid grasp of the foundational concepts in data science, such as statistical methods, data ethics, and the business implications of data analysis.

PC3. Data analysis with Python

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Python Programming: Students should acquire a strong understanding of Python programming, including data structures, control flow, and the use of libraries such as NumPy and Pandas for data manipulation.

KU2. Data Analysis Techniques: A solid foundation in data analysis principles, including data cleaning, transformation, and exploratory data analysis (EDA), is crucial. Students should be knowledgeable about various statistical and visualization techniques used to derive insights from data.

KU3. Data Science Concepts: Beyond the technical aspects, students should gain a conceptual understanding of data science, encompassing ethical considerations, the data science lifecycle, and practical applications of data science in diverse fields, such as healthcare, finance, and marketing.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. • SA1. Fill up documentation applicable to ones role.
• SA2. Read and understand manuals, health and safety instructions, memos, other company documents.
• SA3. Express statements or information clearly so that others can hear and understand.
• SA4. Participate in and understand the main points of simple discussions.

GS2. • SB1. Follow organization rule-based decision making process.
• SB2. Take decision with systematic course of actions and/or response.
• SB3. Planning and organization of work to meet deadlines.
• SB4. Work constructively and collaboratively with others.
• SB5. Follow code of conduct.
• SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
• SB7. Recognize problems and search for solutions.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	**23**	**-**	**-**
PC1. Students should be able to effectively explore and analyze datasets using Python, including data cleaning, feature engineering, and visualization.	20	9	-	-
PC2. Students should exhibit a solid grasp of the foundational concepts in data science, such as statistical methods, data ethics, and the business implications of data analysis.	20	7	-	-
PC3. Data analysis with Python	10	7	-	-
NOS Total	**50**	**23**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1002
NOS Name	CONCEPTUALISING DATA SCIENCE WITH PYTHON
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	3
Credits	2
Version	1.0
Next Review Date	NA

NIE/ITS/N1003: Development of data visualisation and data analysis techniques using Python.

Description

This NOS explores the development of data visualization and analysis techniques using Python, enabling students to create compelling visualizations and extract valuable insights from data sets, enhancing their data-driven decision-making skills.

Scope

The scope covers the following :

- Exploratory Data Analysis (EDA), Interactive Dashboards

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Ability to develop skills in using Python libraries such as Matplotlib, Seaborn, and Plotly to create insightful visualizations that aid in exploring and understanding the characteristics of datasets

PC2. Create dynamic visualizations that allow users to interact with data, filter information, and gain a deeper understanding of trends

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1.
- A.Organizational Context
- KA1.Data-Driven Decision Making
- KA2.Effective Communication
- KA3.Operational Efficiency

KU2.
- B. Technical Knowledge
- KB1.Data Manipulation
- KB2. Visualization Libraries
- KB3.Statistical Analysis

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1.
- SA1.Fill up documentation applicable to ones role. SA2. Read English and/or vernacular language.
- SA3. Read and understand manuals, health and safety instructions, memos, other company documents.
- SA4. Ability to read from different sources- books screens in machines and signage.
- SA5. Understand the various color codes, as per standard elect

GS2. • SB1. Follow organization rule-based decision making process.
• SB2. Take decision with systematic course of actions and/or response.
• The user/individual on the job needs to know and understand how to :
• SB3. Planning and organization of work to meet deadlines.
• SB4. Work constructively and collaboratively with others.
• SB5. Follow code of conduct.
• SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
• SB7. Recognize problems and search for solutions.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	**6**	**-**	**6**
PC1. Ability to develop skills in using Python libraries such as Matplotlib, Seaborn, and Plotly to create insightful visualizations that aid in exploring and understanding the characteristics of datasets	20	3	-	3
PC2. Create dynamic visualizations that allow users to interact with data, filter information, and gain a deeper understanding of trends	20	3	-	3
NOS Total	**40**	**6**	**-**	**6**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1003
NOS Name	Development of data visualisation and data analysis techniques using Python.
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence , Data Science & Engineering
Occupation	AI Development
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

NIE/ITS/N1003: DATA ANALYSIS AND VISUALIZATION

Description

This NOS course provides students with the skills to systematically examine and interpret data through statistical analysis and visualization techniques. Participants learn how to extract meaningful insights, communicate findings effectively, and make data-driven decisions.

Scope

The scope covers the following :

- Data Cleaning and Preparation, Exploratory Data Analysis (EDA), Statistical Analysis, Data Visualization

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Students should demonstrate the ability to clean and preprocess data effectively, removing or handling missing values and outliers, ensuring that data is ready for analysis.

PC2. Performance should be measured based on students' skills in conducting EDA, including their ability to generate meaningful visualizations, compute descriptive statistics, and identify patterns and trends in the data.

PC3. Performance criteria should include the ability to create clear and informative data visualizations that effectively communicate insights from the data.

PC4. Learn New data visualization libraries

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Students should have a deep understanding of data cleaning techniques, data imputation, and the ability to prepare data for analysis by handling missing values, outliers, and inconsistencies.

KU2. Knowledge of various EDA techniques, including summary statistics, data visualization methods, and hypothesis testing, is crucial. Students should understand how to use these tools to gain insights from data.

KU3. A solid foundation in statistical concepts, such as probability, sampling, and hypothesis testing, is essential. Students should comprehend the application of statistical tests and methods to draw meaningful conclusions from data.

KU4. Students should be knowledgeable about data visualization best practices, including understanding chart types, color choices, labeling, and storytelling through data visualization.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. • SA1. Fill up documentation applicable to ones role.
• SA2. Read and understand manuals, health and safety instructions, memos, other company documents.
• SA3. Express statements or information clearly so that others can hear and understand.
• SA4. Participate in and understand the main points of simple discussions.

GS2. • SB1. Follow organization rule-based decision making process.
• SB2. Take decision with systematic course of actions and/or response.
• SB3. Planning and organization of work to meet deadlines.
• SB4. Work constructively and collaboratively with others.
• SB5. Follow code of conduct.
• SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
• SB7. Recognize problems and search for solutions.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	75	**34**	**-**	**-**
PC1. Students should demonstrate the ability to clean and preprocess data effectively, removing or handling missing values and outliers, ensuring that data is ready for analysis.	19	8	-	-
PC2. Performance should be measured based on students' skills in conducting EDA, including their ability to generate meaningful visualizations, compute descriptive statistics, and identify patterns and trends in the data.	19	10	-	-
PC3. Performance criteria should include the ability to create clear and informative data visualizations that effectively communicate insights from the data.	19	8	-	-
PC4. Learn New data visualization libraries	18	8	-	-
NOS Total	**75**	**34**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1003
NOS Name	DATA ANALYSIS AND VISUALIZATION
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	3
Credits	3
Version	1.0
Next Review Date	NA

NIE/ITS/N1004: Develop expertise in data modelling and predictive analysis using Machine Learning.

Description

In this NOS, students will gain advanced proficiency in data modeling and predictive analysis through machine learning techniques, equipping them to harness data-driven predictions and make informed decisions in various domains.

Scope

The scope covers the following :

- Feature Engineering and Selection, Model Evaluation and Validation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Gain proficiency in pre-processing and transforming raw data into suitable formats for modeling. Learn techniques for selecting relevant features that contribute to predictive accuracy while reducing noise.

PC2. Develop expertise in evaluating model performance using metrics like accuracy, precision, recall, F1-score, and others. Learn cross-validation techniques to ensure reliable estimates of a model's performance on unseen data.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1.
- A.Organizational Context
- KA1. Enhanced Decision-Making
- KA2.Optimized Processes
- KA3.Innovation and Product Developmen

KU2.
- B. Technical Knowledge
- KB1.Machine Learning Algorithms
- KB2.Feature Engineering and Selection
- KB3.Model Evaluation and Interpretation

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. • SA1.Fill up documentation applicable to ones role.
• SA2. Read English and/or vernacular language.
• SA3. Read and understand manuals, health and safety instructions, memos, other company documents.
• SA4. Ability to read from different sources- books screens in machines and signage.
• SA5. Understand the various color codes, as per standard electrical,mechanical
• SA6. Express statements or information clearly so that others can hear and understand.
• SA7. Participate in and understand the main points of simple discussions.

GS2. • SB1. Follow organization rule-based decision making process.
• SB2. Take decision with systematic course of actions and/or response.
• The user/individual on the job needs to know and understand how to :
• SB3. Planning and organization of work to meet deadlines.
• SB4. Work constructively and collaboratively with others.
• SB5. Follow code of conduct.
• SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
• SB7. Recognize problems and search for solutions.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	**24**	**-**	**6**
PC1. Gain proficiency in pre-processing and transforming raw data into suitable formats for modeling. Learn techniques for selecting relevant features that contribute to predictive accuracy while reducing noise.	15	12	-	3
PC2. Develop expertise in evaluating model performance using metrics like accuracy, precision, recall, F1-score, and others. Learn cross-validation techniques to ensure reliable estimates of a model's performance on unseen data.	15	12	-	3
NOS Total	**30**	**24**	**-**	**6**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1004
NOS Name	Develop expertise in data modelling and predictive analysis using Machine Learning.
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence , Data Science & Engineering
Occupation	AI Development
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

NIE/ITS/N1004: FUNDAMENTALS OF MACHINE LEARNING

Description

This NOS provides a foundational understanding of machine learning concepts, algorithms, and applications, with a focus on teaching the fundamental principles of how machines can learn from data to make predictions and decisions.

Scope

The scope covers the following :

- Supervised Learning,Unsupervised Learning, Model Evaluation and Optimization

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Students should be able to build machine learning models for various tasks, select appropriate algorithms, and evaluate model performance

PC2. Performance should be measured based on the students' ability to fine-tune model hyperparameters effectively to optimize model performance and generalization.

PC3. Students should demonstrate their problem-solving skills by applying machine learning techniques to real-world problems, interpreting model results, and making informed decisions based on the insights derived from the data.

PC4. Learn new applications and problem solving of machine learning

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Students should have a solid grasp of fundamental machine learning concepts, including supervised and unsupervised learning, feature engineering, model training, and the bias-variance trade-off.

KU2. Knowledge of various machine learning algorithms and techniques, such as linear regression, decision trees, k-means clustering, and neural networks, is essential. Students should understand when and how to apply these algorithms.

KU3. Understanding of model evaluation methods, cross-validation, hyperparameter tuning, and techniques for preventing overfitting or underfitting is critical for building effective machine learning models.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1.
- SA1. Fill up documentation applicable to ones role.
- SA2. Read and understand manuals, health and safety instructions, memos, other company documents.
- SA3. Express statements or information clearly so that others can hear and understand.
- SA4. Participate in and understand the main points of simple discussions.

GS2.
- SB1. Follow organization rule-based decision making process.
- SB2. Take decision with systematic course of actions and/or response.
- SB3. Planning and organization of work to meet deadlines.
- SB4. Work constructively and collaboratively with others.
- SB5. Follow code of conduct.
- SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
- SB7. Recognize problems and search for solutions.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	25	**11**	**30**	**-**
PC1. Students should be able to build machine learning models for various tasks, select appropriate algorithms, and evaluate model performance	7	2	8	-
PC2. Performance should be measured based on the students' ability to fine-tune model hyperparameters effectively to optimize model performance and generalization.	6	2	8	-
PC3. Students should demonstrate their problem-solving skills by applying machine learning techniques to real-world problems, interpreting model results, and making informed decisions based on the insights derived from the data.	7	4	7	-
PC4. Learn new applications and problem solving of machine learning	5	3	7	-
NOS Total	**25**	**11**	**30**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1004
NOS Name	FUNDAMENTALS OF MACHINE LEARNING
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	3
Credits	2
Version	1.0
Next Review Date	NA

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

KU10. how to compute income and expenses
KU11. importance of maintaining safety and security in financial transactions
KU12. different legal rights and laws
KU13. how to operate digital devices and applications safely and securely
KU14. ways to identify business opportunities
KU15. types of customers and their needs
KU16. how to apply for a job and prepare for an interview
KU17. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively using appropriate language
GS2. behave politely and appropriately with all
GS3. perform basic calculations
GS4. solve problems effectively
GS5. be careful and attentive at work
GS6. use time effectively
GS7. maintain hygiene and sanitisation to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**1**	**3**	**-**	**-**
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
Communication Skills	**1**	**1**	**-**	**-**
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**1**	**-**	**-**
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
Financial and Legal Literacy	**3**	**4**	**-**	**-**
PC9. use various financial products and services safely and securely	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
Essential Digital Skills	**4**	**6**	**-**	**-**
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
Entrepreneurship	**3**	**5**	**-**	**-**
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
Customer Service	**2**	**2**	**-**	**-**
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**1**	**3**	**-**	**-**
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

General Assessment Criteria:

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.

2. The assessor assigns roll number

3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical

examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

5. Pass percentage would be 50% marks in each component.

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N1001.PROGRAMMING WITH PYTHON	50	22	-	-	72	20
NIE/ITS/N1002.CONCEPTUALISING DATA SCIENCE WITH PYTHON	50	23	-	-	73	20
NIE/ITS/N1003.DATA ANALYSIS AND VISUALIZATION	75	34	-	-	109	30
NIE/ITS/N1004.FUNDAMENTALS OF MACHINE LEARNING	25	11	30	-	66	21
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	9
Total	**220**	**120**	**30**	**-**	**370**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AI	Artificial Intelligence
IDE	Integrated Development Environment

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.